

Technical Feasibility study

Project: ENR Software Development

Overview

The project works on developing Egyptian National Railway (ENR) system, web app, mobile app, and desktop app, and digital ticketing. The software project provides Assistance services for passengers (I.e. tracking travels, seat selection, Healthcare services, Meals, and provide overview info about each train and its features) to enhance passenger experience.

Technical Requirements

Software Elements And Requirements
LibreOffice(For writing use case Scenarios)
microsoft teams
Bizagi modeler(For creating flowcharts, data flow diagrams, and system architecture diagrams.)
Enterprise Architect(For designing system models using UML standards.)
Figma (Collaborative design tool for interfaces And Prototyping Tool for Technical design)
MySQL Workbench(for designing, visualizing, and managing MySQL databases.)
Visual Paradigm(Supports UML, database diagrams, and more)
Microsoft SQL Server
Firebird(RDBMS for embedded use)
HeidiSQL(for managing SQL databases like MySQL and MariaDB.)
Infrastructure Automation Tools(Terraform)
Database Optimizer Plugin(SolarWinds)

Caching Tool
Encryption Tool(OpenSSL)
Docker(Containerization and Orchestration)
flutter framework
Test Network Infrastructure
Test Server
Jenkins(for creating backups of the database, source code, and configurations)
Jira(for creating backups of the database, source code, and configurations)

Technical expertise and human resources needed

- Security engineering for governmental projects
- Graphic designer
- Terraform(Infrastructure as Code Technology)

Cost Estimation

*Software Technologies and Tools:

- Microsoft 365 Business Standard: \$12.50 per user/month.
- Enterprise Architect (by Sparx Systems) Cost: One-time licensing fees. Corporate Edition: \$299 per user.
- Figma Cost: Subscription-based.
Professional Plan: \$12 per editor/month (billed annually).
- Visual Paradigm Cost: Subscription-based.
Enterprise Edition: \$89 per user/month. The Professional or Enterprise editions are recommended for comprehensive modeling needs.
- Microsoft SQL Server Cost: Enterprise Edition: Approximately \$1500 per core.
- SolarWinds Database Performance Analyzer Cost: Subscription-based, starting at approximately \$900 per database instance.

-Docker Cost: Business: \$21 per user/month.

-Jira Cost: Subscription-based.

Standard Plan: \$7.50 per user/month.

Costs billed monthly, hourly or annually is \$500 considering the design and implementation to take 50 – 60 days.

Total cost is \$2,299

*For Training costs and external expertise resources:

Security engineering for governmental projects \$150 for training with Udemy Enterprise discounts

Terraform \$270 for training with DataComp

Graphic designer : Mr. _____ \$20 per hour

Total cost is \$620

*For Hardware usage:

Resource Name	Cost/Use
Scanner	\$0.50
Webserver	\$10.00
PC	\$0.50
Laptop	\$0.50

Total cost approximately is \$50

Considerations and possible risks

Tools like Microsoft SQL Server, Docker, and Jenkins may require robust server infrastructure, leading to additional hardware or cloud service expenses.

Jenkins is Free, but infrastructure and maintenance costs apply if self-hosted.

Enterprise Architect (by Sparx Systems) for additional modules may require higher costs can't be pre-expected

Conclusion

Technical costs is estimated to be a total of \$3,200, Risks like possible scope creep, limitations of certain tools or potential extended duration can increase costs up to 15%.

Revenues is estimated to be \$5000 in the first 30 days of launching if the number of users is increased by only 5% (our goal is to reach 15% by march 2025).

Moreover, return on investment (ROI) on the project includes enhancing the company's reputation and credibility, which strengthens brand recognition and builds trust among potential clients. also creates a competitive advantage

Moreover, successful government collaborations lead to long-term partnerships and networking opportunities. also, Employees and developers will benefit from skill enhancement and training for new technologies with larger scales.

To sum up, The project implementation technically is within the proposed budget and can be executed without financial barriers